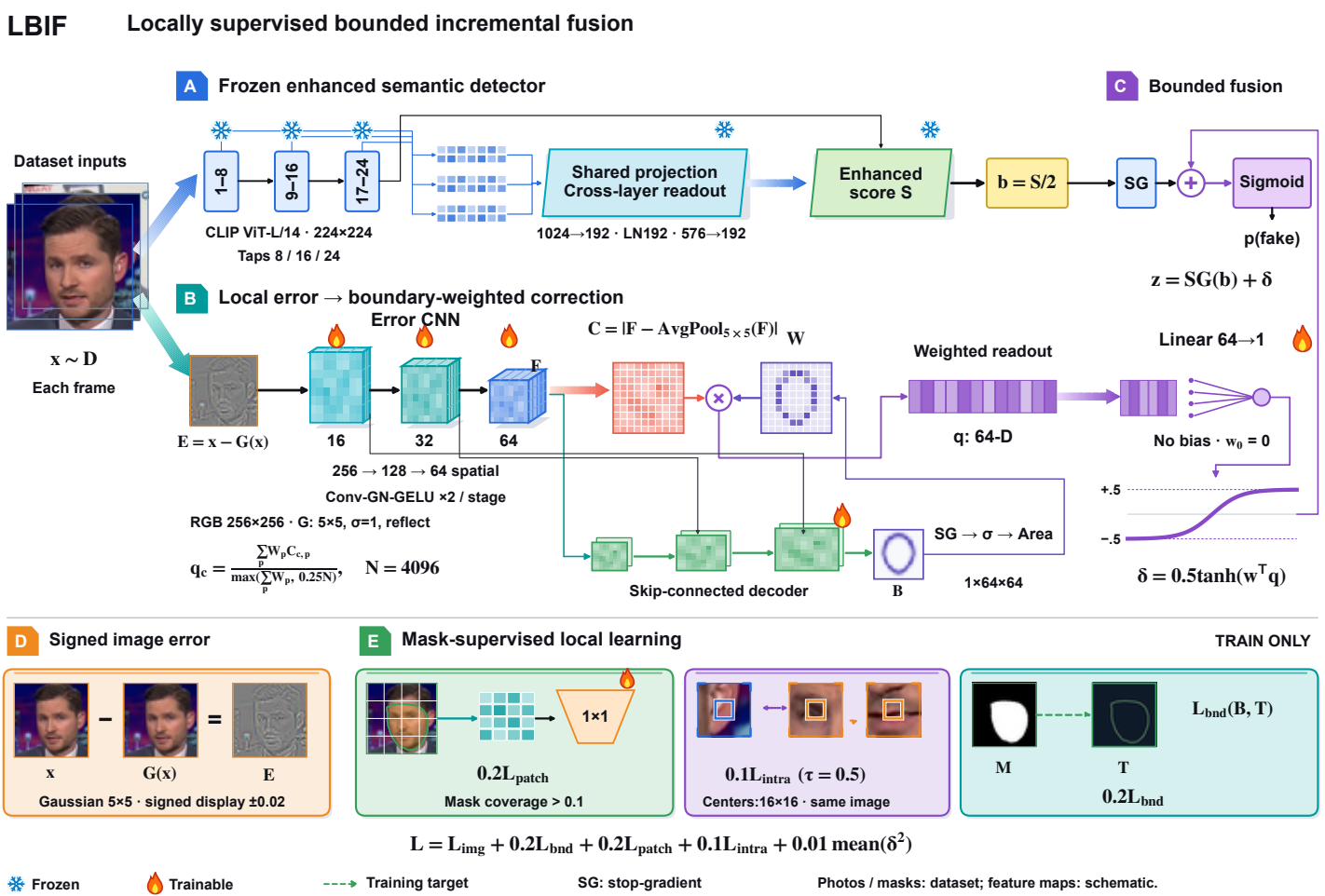
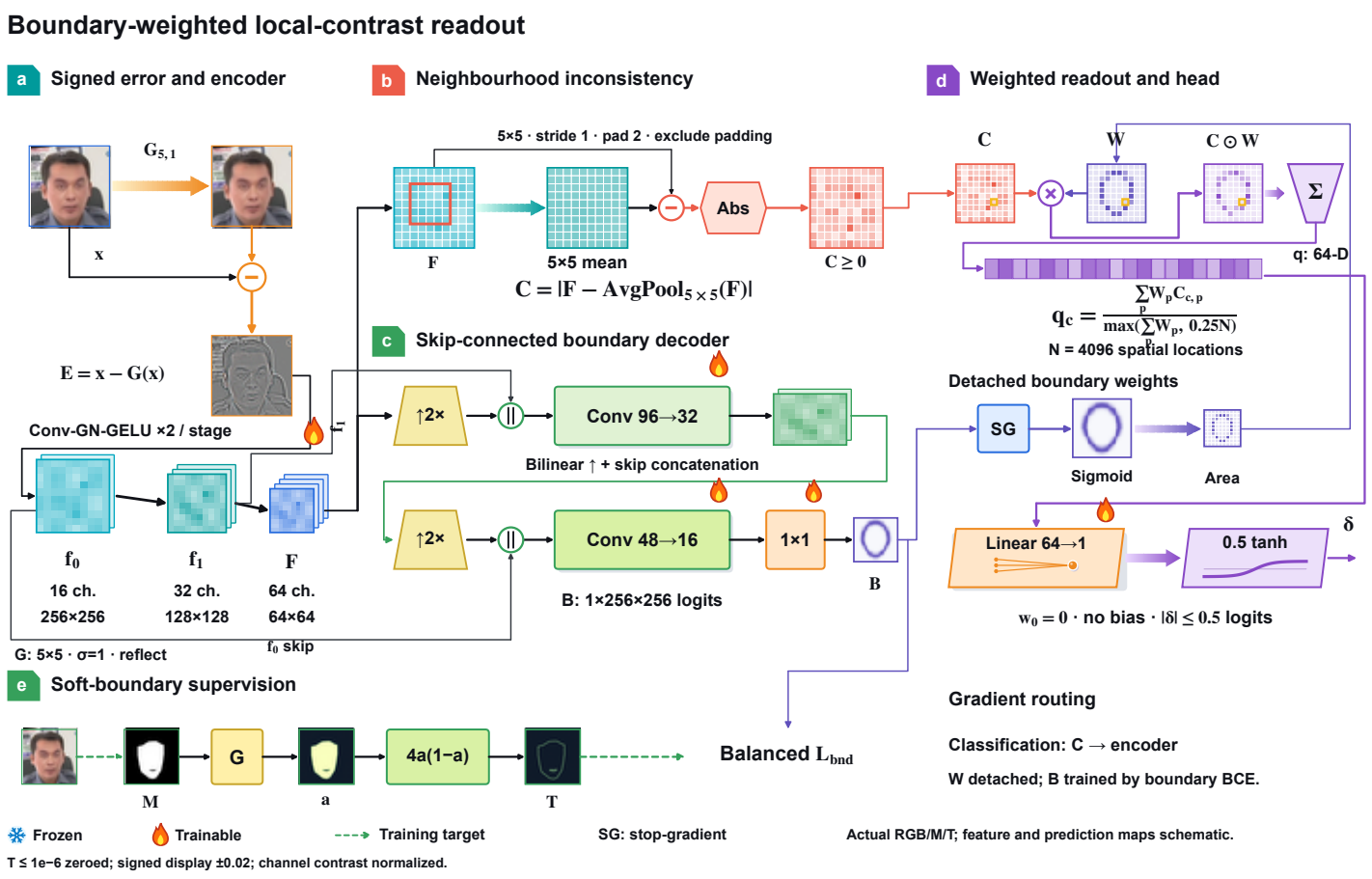

LBIF
Locally supervised bounded incremental fusion
A
Frozen enhanced semantic detector
Dataset inputs
1–8
9–16
17–24
CLIP ViT-L/14 · 224×224
Taps 8 / 16 / 24
Shared projection
Cross-layer readout
1024→192 · LN192 · 576→192
Enhanced score S
b = S/2
SG
Sigmoid
p(fake)
C
Bounded fusion
z = SG(b) + δ
B
Local error → boundary-weighted correction
Error CNN
x ∼ D
Each frame
E = x − G(x)
16
32
64
F
256 → 128 → 64 spatial
Conv-GN-GELU ×2 / stage
RGB 256×256 · G: 5×5, σ=1, reflect
C = |F − AvgPool5×5(F)|
W
Weighted readout
q: 64-D
Linear 64→1
No bias · w0 = 0
+.5
−.5
δ = 0.5tanh(wTq)
Skip-connected decoder
B
SG → σ → Area
1×64×64
D
Signed image error
x
G(x)
E
Gaussian 5×5 · signed display ±0.02
E
Mask-supervised local learning
TRAIN ONLY
1×1
0.2Lpatch
Mask coverage > 0.1
0.1Lintra (τ = 0.5)
Centers:16×16 · same image
M
T
Lbnd(B, T)
0.2Lbnd
L = Limg + 0.2Lbnd + 0.2Lpatch + 0.1Lintra + 0.01 mean(δ2)
Frozen
Trainable
Training target
SG: stop-gradient
Photos / masks: dataset; feature maps: schematic.

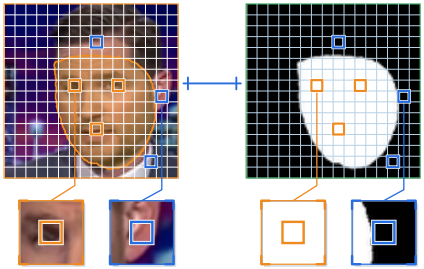

Boundary-weighted local-contrast readout
a
Signed error and encoder
b
Neighbourhood inconsistency
d
Weighted readout and head
$G_{5,1}$
x
$E = x - G(x)$
Conv-GN-GELU ×2 / stage
f_0
16 ch.
256×256
f_1
32 ch.
128×128
F
64 ch.
64×64
f_1
f_0 skip
G: 5×5 · σ=1 · reflect
5×5 · stride 1 · pad 2 · exclude padding
F
5×5 mean
Abs
$C \geq 0$
$C = |F - AvgPool_{5\times5}(F)|$
c
Skip-connected boundary decoder
↑2×
Conv 96→32
Bilinear ↑ + skip concatenation
↑2×
Conv 48→16
1×1
B
B: 1×256×256 logits
C
W
C ⊙ W
Σ
q: 64-D
$q_c = \frac{\sum_p W_p C_{c,p}}{\max(\sum_p W_p,\ 0.25N)}$
N = 4096 spatial locations
Detached boundary weights
SG
Sigmoid
Area
Linear 64→1
0.5 tanh
δ
$w_0 = 0$ · no bias · $|\delta| \leq 0.5$ logits
e
Soft-boundary supervision
M
G
a
4a(1−a)
T
Balanced L_{bnd}
Gradient routing
Classification: C → encoder
W detached; B trained by boundary BCE.
Frozen
Trainable
Training target
SG: stop-gradient
Actual RGB/M/T; feature and prediction maps schematic.
T ≤ 1e−6 zeroed; signed display ±0.02; channel contrast normalized.

Mask-guided patch learning: from pixels to local relations

a Region labels from an aligned mask

Aligned RGB

Paired mask M



16×16 grid · zooms show boxed 16×16 centers

$$t_j = 1[\text{mean}(M_j) > 0.1]$$

4 × 4 numerical example (as in draft)

0	0.38	1	1
0	0.71	1	1
0.04	0.97	1	1
0.03	0.99	1	1

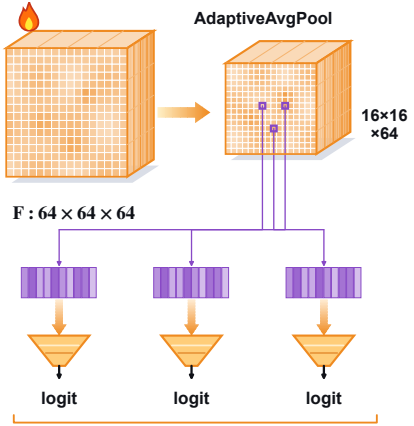
Coverage

0	1	1	1
0	1	1	1
0	1	1	1
0	1	1	1

Region labels

Photo-mask and numeric toy are separate examples.

b One CNN, 256 contextual vectors



Class-balanced within image; then image mean.

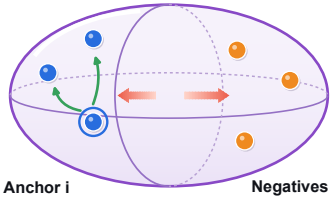
c Contrast within one fake image

One valid fake image v



Context zooms; boxed centers = 16×16

L2-normalized 64-D vectors



Same class: pull together

Different classes: push apart

$$L_{intra} \quad (\tau = 0.5)$$

No self-pairs · within-image pairs only

TRAIN ONLY

Valid masks

Contrast: ≥2 patches / class

Patch head removed at test

Actual RGB / M; context centers are real 16×16 cells. Feature tensors and unit-sphere relations are schematic.